UNIT-II

Corrosion Science

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Corrosion is Extractive Metallurgy in reverse

EXTRACTIVE METALLURGY

ORE

Stable Natural combined form

Lower Energy State

Unstable pure form

Higher Energy Excited State

METAL

CORROSION

Corrosion:

Any process of ***destruction (or deterioration)*** and consequent loss of a solid metallic material, through an unwanted ***chemical*** or ***electrochemical*** attack by its ***environment***, starting at its ***surface***.

Examples:-

i) Rusting of Iron

ii) **Copper corrosion**

iii) **Corrosion of Silver**

BEFORE AFTER TARNISH

Two types of Corrosion:

1) **Dry** or **Chemical** Corrosion:- Involves the direct attack of metals by dry gases through chemical reactions.

2) **Wet** or **Electrochemical** Corrosion :- Involves electrochemical reaction of metal with moisture and oxygen present in the atmosphere.

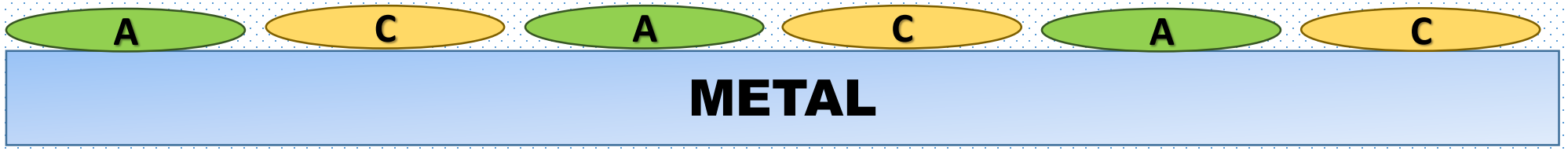
Electrochemical Theory of Corrosion taking Fe as example

Electrochemical corrosion occurs ***when a metal is in contact with a corrosive environment***.

1) *Formation of a large number of minute galvanic cells (anodic and cathodic areas) on the metal surface.*
2) *At anode, the metal undergoes oxidation (corrosion takes place) & electrons are liberated.*

$$M \rightarrow M^{n+} + ne^-$$

Taking Fe as an example, $\mathbf{Fe \rightarrow Fe^{2+} + 2e^-}$



3) At cathode, **reduction** occurs depending on the environmental conditions.

	HYDROGEN EVOLUTION In the absence of Oxygen (O_2-deficient environment)	**OXYGEN ABSORPTION** In the presence of Oxygen (O_2-rich environment)
Acidic	$2H^+ + 2e^- \rightarrow H_2\uparrow$	$2H^+ + \frac{1}{2} O_2 + 2e^- \rightarrow H_2O$
Alkaline/ Neutral	$2H_2O + 2e^- \rightarrow H_2\uparrow + 2OH^-$	$H_2O + \frac{1}{2} O_2 + 2e^- \rightarrow 2OH^-$

Rust formation

Fe^{2+} (produced at the anode) and OH^- (produced at the cathode) diffuse towards each other and form iron hydroxide on the metal surface.

$$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$$

O_2-rich environment $\quad 2Fe(OH)_2 + \frac{1}{2} O_2 + H_2O \rightarrow Fe_2O_3.3H_2O$ — Hydrated ferric oxide- Yellow rust

O_2-deficient environment $\quad 3Fe(OH)_2 + \frac{1}{2} O_2 \rightarrow Fe_3O_4.3H_2O$ — Magnetic oxide- Black rust

Types of Corrosion

1) Differential Metal Corrosion (Galvanic Corrosion)

- Arises when *two* dissimilar metals are in contact with each other in the same corrosive environment.
- The one with *lower* electrode potential behaves as *anode* (undergoes corrosion).
- The one with *higher* electrode potential behaves as *cathode* (remains passive).

Cause/ Driving force for corrosion:- Potential difference between the two dissimilar metals.

Mechanism: ***At anode***- **$M \rightarrow Mn^{+} + ne^{-}$** (less noble metal)

At cathode-

Acidic medium-**(a) $2H^{+} + 2e^{-} \rightarrow H_2\uparrow$**

Alkaline/Neutral medium- **(b) $H_2O + ½\ O_2 + 2e- \rightarrow 2OH-$**

Examples:

- Steel screws in a brass marine hardware.
- Lead antimony solder around the copper wire
- Steel pipe connected to copper plumbing.

Galvanic Series

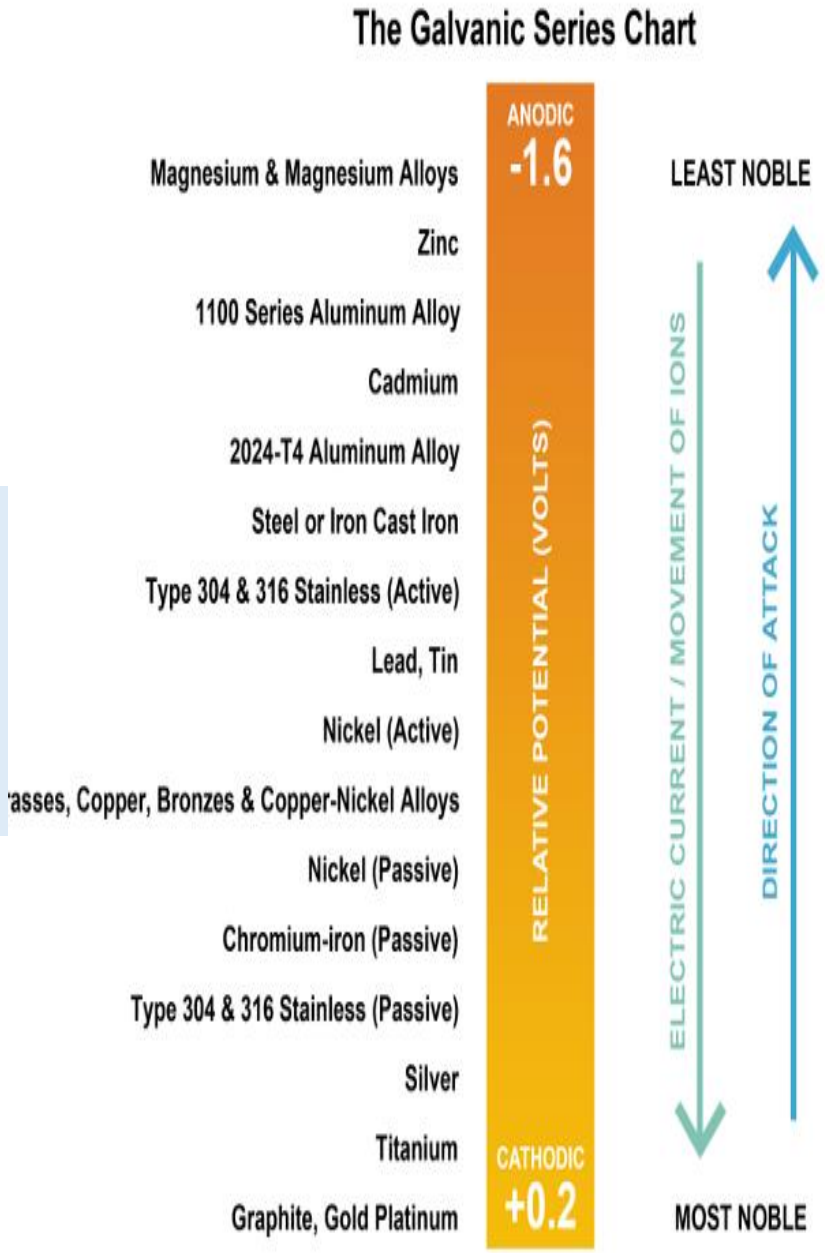
Galvanic couple: Li/Mg, Zn/Fe, Fe/Sn, Fe/Cu, Zn/Sn, Zn/Cu, Cu/ Ag

Exception: Ti-Ag and Al-Zn couple

Ti and Al form a ***stable*** product on corrosion i.e. TiO_2 and Al_2O_3 respectively.

This forms a ***protective film*** on the metal surface ***not*** allowing further corrosion.

Arrangement of ***metals*** and ***alloys*** in the increasing order of their ***corrosion resistance*** *in the given environment* is called as **Galvanic Series.**



2) Differential Aeration Corrosion

- Arises when the same metal is exposed to different oxygen concentrations.
- The part of the metal exposed to lower concentration of oxygen behaves as anode (undergoes corrosion).
- The part of the metal exposed to higher concentration of oxygen behaves as cathode (remains passive).

At the anode $M \rightarrow M^{n+} + ne$

At the cathode $H_2O + \frac{1}{2} O_2 + 2e^- \rightarrow 2OH^-$

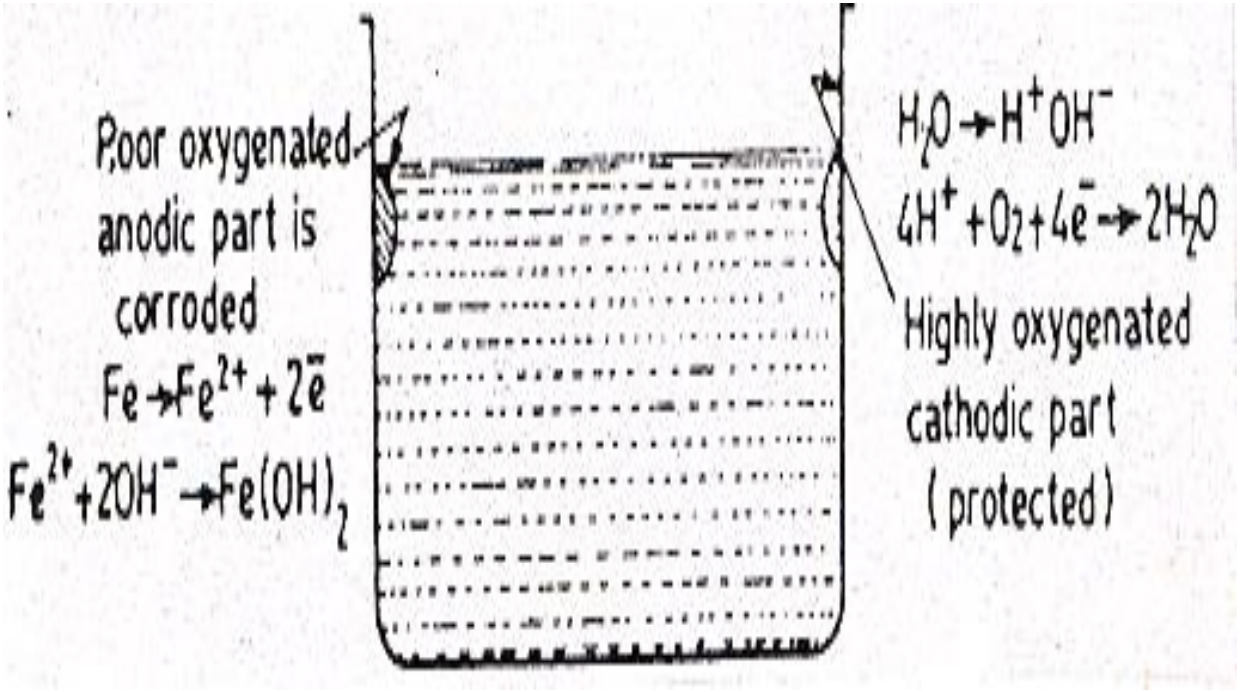
Definition:- *Corrosion of metals arising as result of the formation of oxygen concentration cells due to the uneven supply of air on the metal surface.*

Types

- Waterline corrosion
- Pitting corrosion

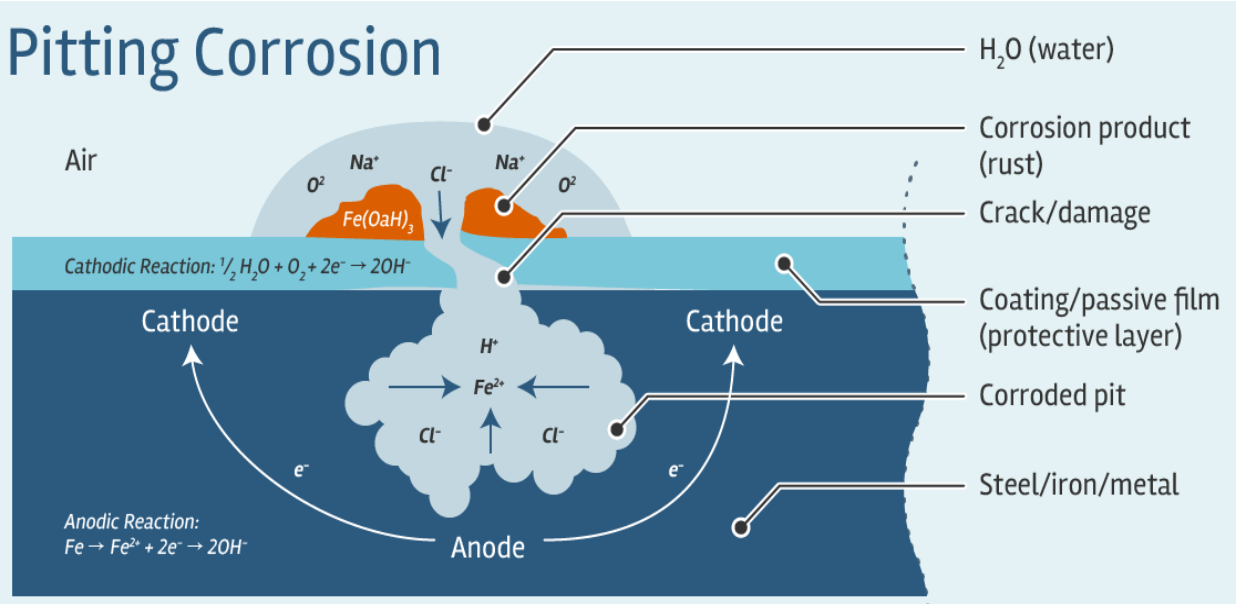
i) Waterline Corrosion

- Occurs when a part of the metal is subjected to water.
- The part of the metal in contact with water (lower O_2 concentration) behaves as ***anode***.
- The part of the metal in contact with atmosphere (higher O_2 concentration) behaves as ***cathode***.

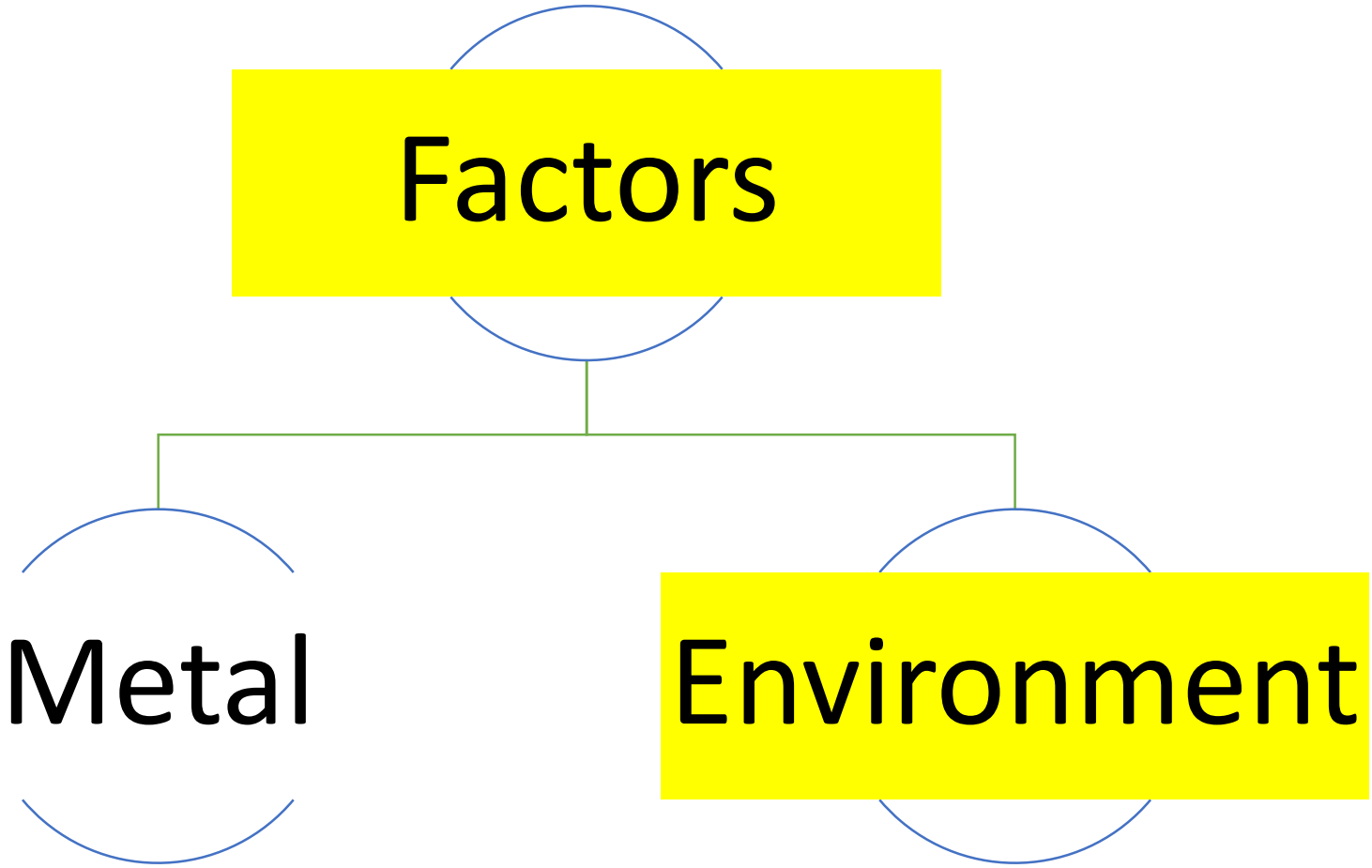


ii) Pitting Corrosion

- Occurs in **pits** formed due to deposition of impurity, moisture on metal surface.
- The part of the metal under the impurity is exposed to **lesser** oxygen and hence behaves as **anode** and undergoes corrosion.
- The part of the metal free of impurity is exposed to atmospheric oxygen and hence behaves as **cathode**.
- **Localised** form of corrosion
- Very dangerous as remains undetected.



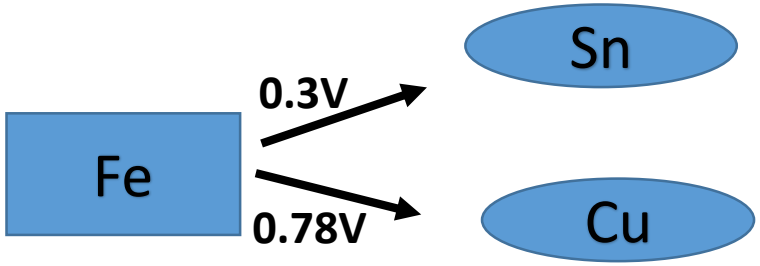
Factors affecting rate of corrosion



Factors related to nature of the Metal

1) Electrode Potential

- When two different metals are in contact in the same atmosphere → **galvanic corrosion occurs.**
- The potential difference(**Open Circuit Potential Difference**) between anodic and cathodic areas controls corrosion rate.
- ***Greater potential difference(OCPD) → Higher corrosion rate.***



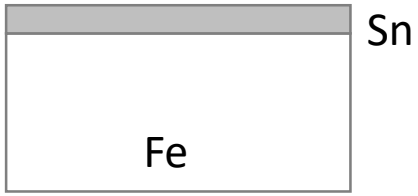
The ***farther*** the positions of the metals in the galvanic series, ***greater*** is the intensity of corrosion on the anodic metal.

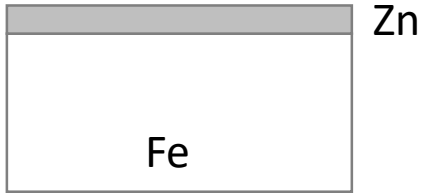
- *Avoid using dissimilar metals together whenever possible.*
- *Use same metal/alloy for components like: Bolt & nut, Screw & washer*

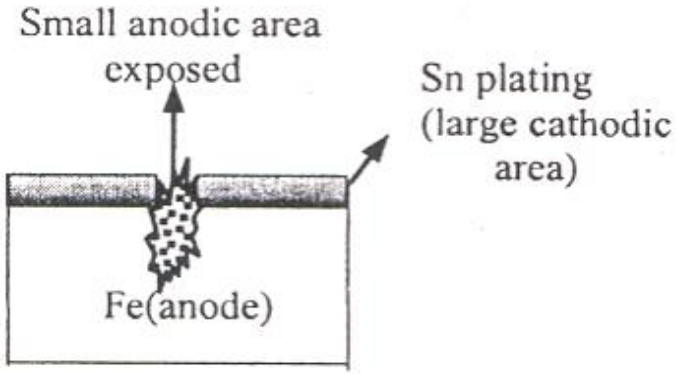
Factors related to nature of the Metal

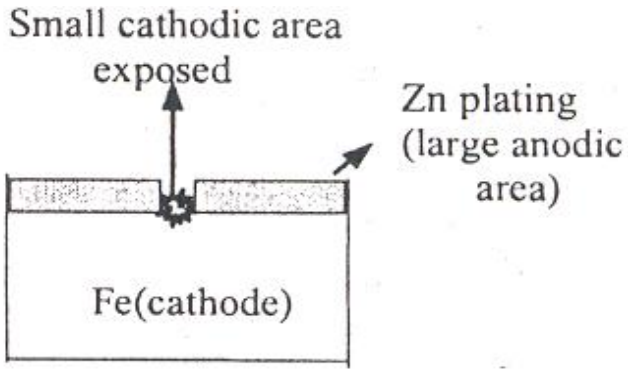
2) Relative areas of the anode and cathode

$$Rate\ of\ corroson \propto \frac{cathodic\ area}{anodic\ area}$$









Factors related to nature of the Metal

3) Nature of corrosion product and tendency to form protective film:

Al, Ti, Cr

PASSIVE

- ✓ Insoluble
- ✓ Stable
- ✓ Non-porous
- ✓ Uniform

Protective Film

Fe, Zn, Mg

ACTIVE

- ✓ Soluble
- ✓ Unstable
- ✓ Porous
- ✓ Non-uniform

~~Protective~~ Film

Factors related to nature of the Metal

4) Hydrogen Overvoltage

Low Hydrogen Overvoltage → **Easy** liberation of H_2 gas → **Increases** cathodic reaction rate → **Increases** anodic reaction rate → Metal corrodes **faster**

High Hydrogen Overvoltage- Lower corrosion rate

Factors related to nature of the Environment

1) pH of the medium

pH<3	• Rate of corrosion is **high** because of high conc. of H^+
pH>10	• Rate of corrosion is **low** because of formation of hydroxide film
pH 3-10	• Rate of corrosion **increases** with the **availability of O_2**

Ships submerged under sea water undergo slow corrosion. WHY???

Factors related to nature of the Environment

2) Temperature

Rise in temperature → Conductance increases → Reactions occur faster → Rate of corrosion **increases**

Factors related to nature of the Environment

3) Humidity

- Presence of *water* (moisture) makes the atmosphere humid.
- Water acts like an *electrolyte* leading to corrosion.

4) Presence of Impurities

- Impurities (dissolved gases and acids) can *decrease the pH* of the atmosphere leading to corrosion.
- Impurities (NaCl) can also *improve conductivity* of the environment leading to corrosion.

Corrosion Control

- Protective Coatings
  - Metal
    - Anodic (Galvanization)
    - Cathodic (Tinning)
  - Inorganic
    - Anodizing
    - [redacted]
- Cathodic Protection
  - Sacrificial Anode
  - [redacted]

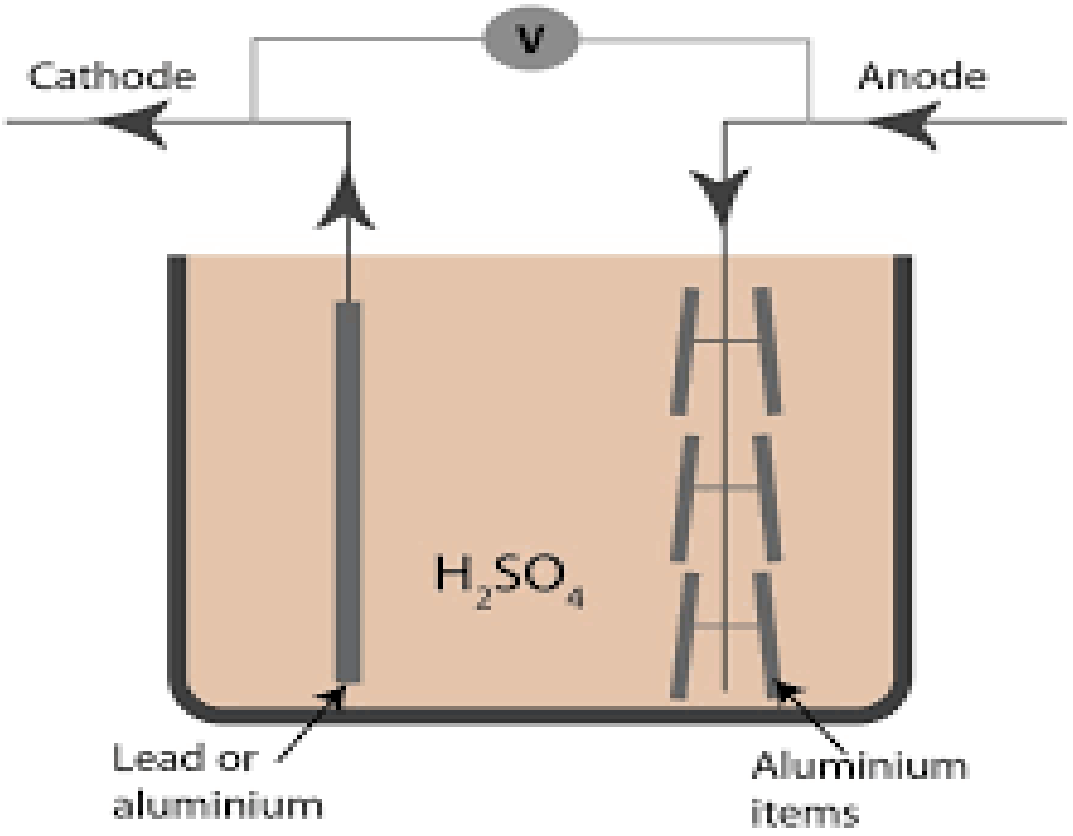
Protective Inorganic Coating

- Chemical conversion coatings
- Surface layer of base metal converted to its ***inorganic*** compound
- The inorganic coating forms a ***protective film***

ANODISING

- Generally produced on non-ferrous metals like **Zn, Al, Mg** and their alloys by **anodic oxidation process.**
- It is carried out by passing a moderate direct current through a bath in which **base metal** is made as ***anode***.
- The bath usually contains sulphuric acid, chromic acid, phosphoric acid, oxalic acid or boric acid.

Aluminium coating



Experimental conditions:

- Bath – Acidic
- Temp - 35-40°C
- CD – 10-20 mA/cm^2

Post-treatment:

- Immerse it in hot boiling water
- The coating becomes hydrated and non-porous.

Metal Coating

Deposition of a protective metal on the base metal.

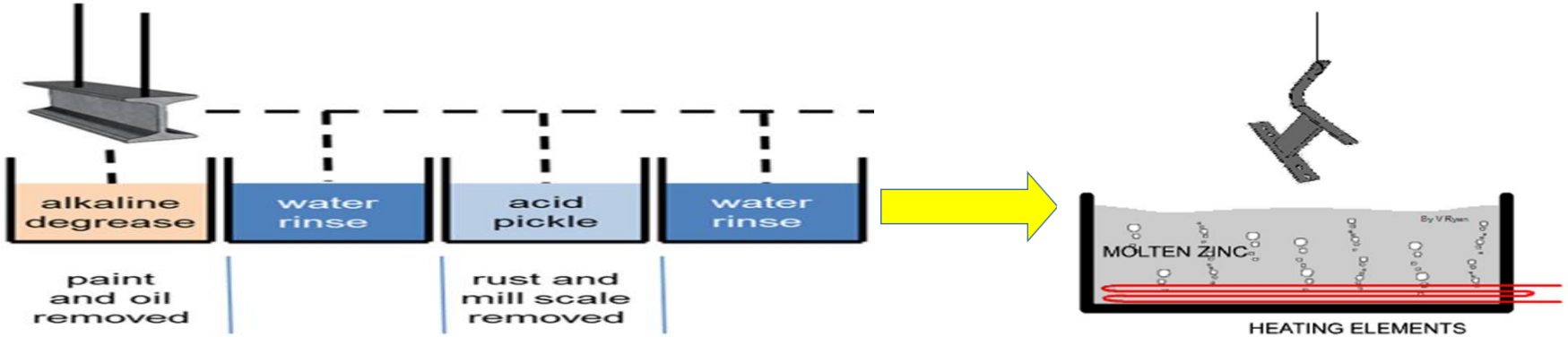
i) Anodic Metal Coating

- Coating base metal with ***anodic*** metals.
- Example: Zn (Galvanization), Mg
- Also called ***Sacrificial Coating*** as though the coating peels off, the base metal remains ***protected***.
- **Uses:** roofing sheets, pipes, tubes, nails,buckets etc.

ii) Cathodic Metal Coating

- Coating base metal with ***cathodic*** metals.
- Example: Sn (Tinning), Ag
- If the coating peels off, ***intense localized corrosion*** of base metal takes place. Hence the coating ***must be continuous***.
- **Uses:** food storing containers, food packaging, refrigeration components, etc.

Galvanization

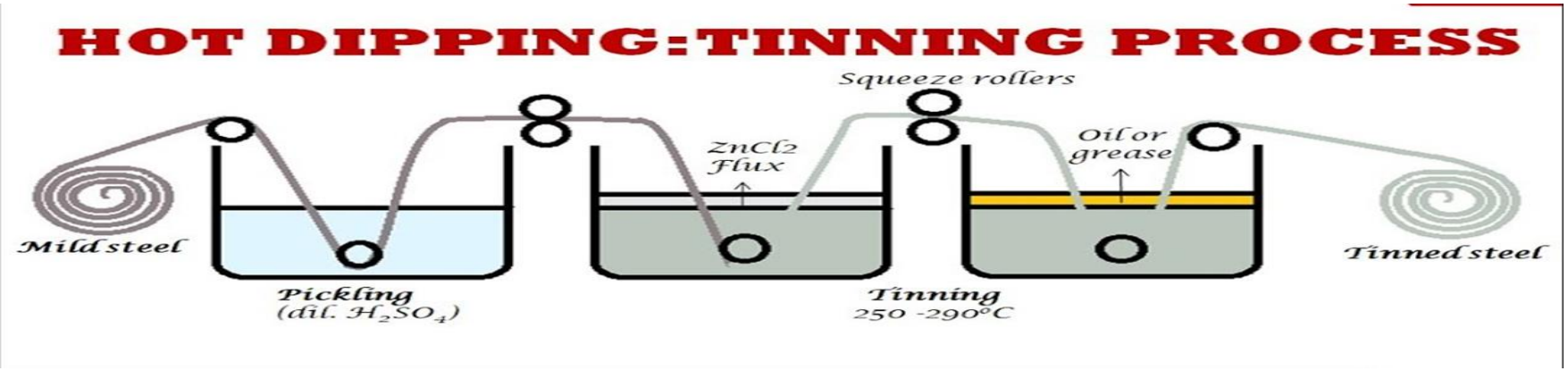


Process of coating base metal surface with zinc metal to prevent corrosion – Hot dipping method

- The metal surface is first washed with *organic solvents* to remove oil and grease deposits.
- Rust and other deposits are removed by washing with *dilute sulphuric acid (pickling)*.
- Finally the article is washed with *water* and air dried.
- The article is then dipped in a bath of *molten zinc*, maintained at *425-430^{0}C* and covered with a *flux of ammonium chloride* to prevent the oxidation of molten zinc.
- The superfluous (excess) zinc on the surface is removed by passing through a pair of *hot rollers*, which wipes out excess of zinc coating and produces a thin coating.

Tinning

Process of coating base metal surface with Tin.

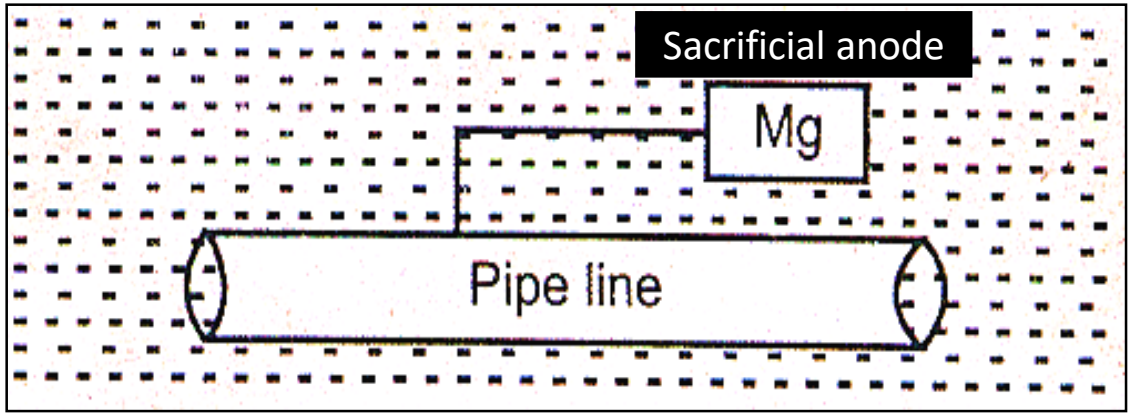


- The metal surface is first washed with *organic solvents* to remove oil and grease deposits.
- Rust and other deposits are removed by washing with *dilute sulphuric acid (pickling)*.
- Finally the article is washed with *water* and air dried.
- The clean and dry sheet is passed through molten *zinc chloride flux*. The flux helps in molten metal to adhere on the metal surface.
- It is then passed through a tank that contains *molten tin (250 - 290° C)*.
- Finally passed through a series of *rollers* immersed in palm oil. The rollers wipe out excess tin deposit and produce a continuous, thin coating of tin on the sheet. The oil prevents the oxidation of the tin coated surface.

Cathodic Protection

Protection of base metal by completely ***converting it into a cathode****.*

Sacrificial Anode Method



Sacrificial anode metals:- Mg, Al, Zn and their alloys.

Applications:-

Protection of buried pipelines, underground cables, ship- hulls, water tanks, etc.

Thank You